PAPER_670: UQFF Black Hole Accretion Model

Subtitle: Bondi accretion with UQFF vacuum field corrections: aether density boost, f_TRZ enhancement, and U_m impedance. M(t) evolution. **Module:** UQFFBlackHoleAccretionModel
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Abstract

We derive a UQFF-modified Bondi accretion rate and simulate black hole mass evolution M(t). UQFF aether density adds an effective contribution to the ambient density, while f_TRZ and U_m further modulate the accretion flow.

1. Standard Bondi Accretion

$$\dot{M}_{Bondi} = 4\pi\lambda_B \frac{(GM)^2\rho_\infty}{c_s^3}$$

$c_s = \sqrt{\gamma_{ad} k_B T_\infty / m_p}$, $\lambda_B = 1/4$ (adiabatic, γ=5/3).

2. UQFF Modifications

Effective Density

$$\rho_{eff} = \rho_\infty + \rho_{UA} - \rho_{SCm}$$

S_TRZ Boost

$$S_{TRZ} = 1 + f_{TRZ} = 1.1$$

U_m Impedance

$$S_{U_m} = 1 - \exp\left(-\frac{U_m}{k_B T_\infty}\right)$$

3. UQFF Accretion Rate

$$\dot{M}_{UQFF} = \dot{M}_{Bondi} \cdot \frac{\rho_{eff}}{\rho_\infty} \cdot S_{TRZ} \cdot S_{U_m}$$

4. Eddington Limit

$$L_{Edd} = \frac{4\pi G M m_p c}{\sigma_T}$$

$$\dot{M}_{Edd} = \frac{L_{Edd}}{\eta c^2}, \quad \eta = 0.1$$

5. M(t) Evolution

$$\frac{dM}{dt} = \dot{M}_{UQFF} - \frac{L_{UQFF}}{c^2}$$

Euler integration; Sgr A* context: M(t) nearly constant (evasion of both Hawking and super-Eddington accretion).

6. C++ Module

`UQFFBlackHoleAccretionModel.h / .cpp` — Session 172 CP4 #254 — `UQFFBlackHoleAccretionModelCalculator`

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